

ENTERPRISE AI GOVERNANCE SYSTEMS ENGINEERING MANUAL

End to End Audit Lifecycle and Continuous Control Monitoring Specification

Document Purpose: This manual establishes an integrated technical framework for AI lifecycle compliance. It outlines the transition into an automated enterprise Cyber GRC architecture by wiring up vendor risk, automated incident generation, and control expiration tracking into a single relational engine.

EXECUTIVE OVERVIEW

System Purpose

This system bridges the gap between static compliance frameworks and live engineering environments. It provides a central operating picture for enterprise artificial intelligence governance, ensuring all models, datasets, and third party vendors are continuously aligned with federal guidelines like the NIST AI Risk Management Framework.

Operational Mechanics

The architecture utilizes an event driven pipeline to ingest live telemetry from model testing environments. It applies autonomous artificial intelligence gatekeepers to validate structural schema at the boundary, normalizes the data, and commits compliant logs directly into an immutable evidence vault. It entirely replaces manual evidence collection with automated telemetry routing.

Strategic Business Value

Unlike legacy compliance audits that decay the moment they are recorded, this cyber governance architecture provides continuous control monitoring. By automating evidence collection and mapping it directly to a relational risk register, this system guarantees real time visibility, enforces strict non repudiation, and automatically routes compliance failures into active incident response queues before they become systemic vulnerabilities.

PHASE 1: FOUNDATION AND SUPPLY CHAIN

Asset Intake and Classification

Enterprise AI Governance System								
Data Automations Interfaces Forms								
AI Asset Inventory								
NIST AI RMF Core Evidence Vault Risk Register Vendor Directory Audit Findings (CAPA) API Staging - Scan Ingestion								
Grid view Hide fields Filter Group Sort Color Share and sync								
	System ID	System Name	Business Unit	Risk Tier	Lifecycle Stage	Model Owner	Technical Stack	Data Sensitivity
1	AI-001	Customer Insights Engine	Marketing	High	Production	Joseph Morrow	Python, TensorFlow, AWS S...	PII
2	AI-002	Fraud Detection Model	Finance	Critical	Production	Joseph Morrow	Scala, Spark MLlib, Azure ML	Financial
3	AI-003	Churn Prediction	Customer Success	Medium	Pilot		Python, scikit-learn, GCP AI...	Internal
4	AI-004	Document Summarizer	Legal	Low	Development		Python, HuggingFace Trans...	Confidential
5	AI-005	Sales Forecasting	Sales	Medium	Production		R, Prophet, AWS Lambda	Internal
6	AI-006	Chatbot Assistant	Support	High	Production		Node.js, Dialogflow, GCP	PII
7	AI-007	Inventory Optimizer	Supply Chain	Medium	Pilot		Python, PyTorch, Azure ML	Internal
8	AI-008	Resume Screener	HR	High	Production		Python, spaCy, AWS Sagem...	PII
9	AI-009	Sentiment Analyzer	Communications	Low	Development		Python, NLTK, Flask	Public
10	AI-010	Anomaly Detector	IT Security	Critical	Production		Go, TensorFlow, Kubernetes	Sensitive
11	AI-011	Personalization Engine	E-commerce	High	Production		Python, Keras, AWS EC2	PII
12	AI-012	Pricing Optimizer	Product	Medium	Pilot		R, Shiny, GCP	Internal
13	AI-013	Voice Recognition	Operations	High	Production		C++, Kaldi, AWS Lambda	PII
14	AI-014	Image Classifier	R&D	Low	Development		Python, OpenCV, PyTorch	Public
15	AI-015	Translation Service	Globalization	Medium	Production		Java, TensorFlow, Azure Fu...	Internal

Figure 1.0: AI Asset Inventory. The system establishes visibility over all enterprise AI deployments. Models are systematically inventoried, assigned explicit domain ownership, and evaluated across inherent risk metrics and technical dimensions.

Third Party Vendor Risk

Enterprise AI Governance System

Data

Automations

Interfaces

Forms

🔄

Launch

🔗

Share

AI Asset Inventory

NIST AI RMF Core

Evidence Vault

Risk Register

Vendor Directory

Audit Findings (CAPA)

API Staging - Scan Ingestion

Tools

☰

Grid view

1 hidden field

Filter

Group

Sort

Color

Share and sync

☐	A Vendor Name	Service Type	SOC 2 Type II Status	Expiration Date	Data Processing...	Vendor Risk Tier	+
1	Amazon Web Services	Infrastructure	Valid		👍	Critical	
2	Microsoft Azure	Infrastructure	Valid		👍	High	
3	Google Cloud Platform	Infrastructure	Valid		👍	High	
4	OpenAI	API	Not Provided		👍	Medium	
5	Anthropic	API	Not Provided		👍	Medium	
6	IBM Watson	SaaS	Valid		👍	Medium	
7	Databricks	SaaS	Valid		👍	High	
8	Hugging Face	API	Not Provided		👍	Low	
9	Salesforce Einstein	SaaS	Valid		👍	Medium	
10	Cohere	API	Not Provided		👍	Low	
+							

Figure 2.0: Vendor Directory. External dependencies are mapped directly to internal assets. The architecture tracks SOC 2 Type II Status, Data Processing Agreements, and associated Vendor Risk Tiers.

Regulatory Hardcoding

Enterprise AI Governance System

Data

Automations

Interfaces

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Launch

Share

AI Asset Inventory

NIST AI RMF Core

Evidence Vault

Risk Register

Vendor Directory

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API Staging - Scan Ingestion

+

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Grid view

Hide fields

Filter

Group

Sort

Color

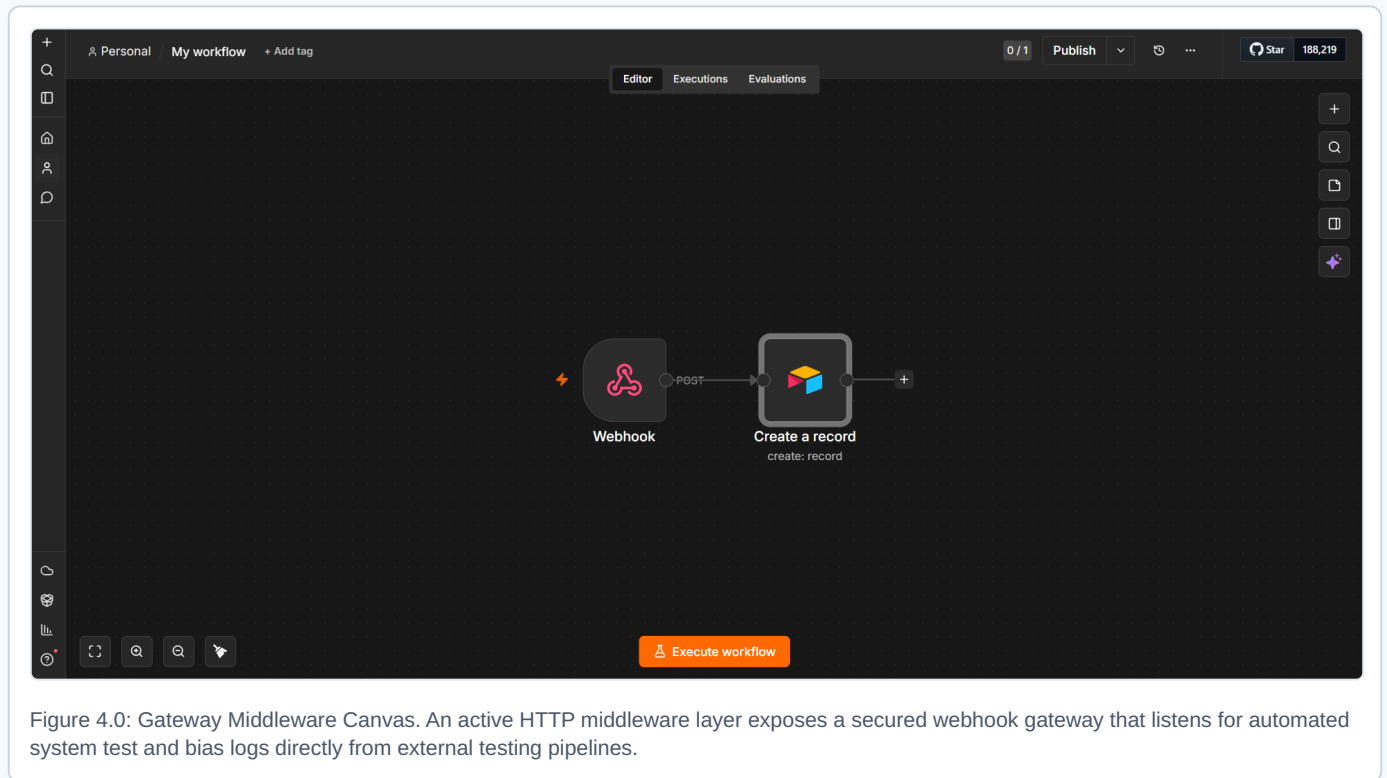
Share and sync

	A Control ID	Function	A Category	NIST Outcome...	Audit Requirement	Reference URL	Evidence Vault	
	GOV-1.1	Govern	Governance Policies	Establish and maintain AI g...	Documented governance p...	https://www.nist.gov/ai-ris...	AI-001 >> GOV-1.1	
2	GOV-2.2	Govern	Risk Management Roles	Assign and communicate r...	Roles and responsibilities a...	https://www.nist.gov/ai-ris...	AI-002 >> GOV-2.2	
3	MAP-1.3	Map	Contextual Understanding	Identify and document the ...	Context and use cases are ...	https://www.nist.gov/ai-ris...	AI-003 >> MAP-1.3	
4	MAP-2.1	Map	Stakeholder Engagement	Engage stakeholders to ide...	Stakeholder engagement s...	https://www.nist.gov/ai-ris...	AI-004 >> MAP-2.1	
5	MEAS-1.1	Measure	Performance Metrics	Define and monitor perfor...	Performance metrics are tr...	https://www.nist.gov/ai-ris...	AI-005 >> MEAS-1.1	
6	MEAS-2.2	Measure	Risk Assessment	Conduct regular risk assess...	Risk assessments are perfor...	https://www.nist.gov/ai-ris...	AI-006 >> MEAS-2.2	
7	MAN-1.1	Manage	Incident Response	Develop and implement in...	Incident response plans are...	https://www.nist.gov/ai-ris...	AI-007 >> MAN-1.1	
8	MAN-2.3	Manage	Continuous Improvement	Establish processes for con...	Continuous improvement a...	https://www.nist.gov/ai-ris...	AI-008 >> MAN-2.3	
9	GOV-3.1	Govern	Ethical Principles	Integrate ethical principles ...	Ethical guidelines are refer...	https://www.nist.gov/ai-ris...	AI-009 >> GOV-3.1	
10	MAP-3.2	Map	Data Mapping	Map and document data s...	Data flow diagrams are mai...	https://www.nist.gov/ai-ris...	AI-010 >> MAP-3.2	
11	MEAS-3.1	Measure	Bias Detection	Implement processes to de...	Bias detection results are r...	https://www.nist.gov/ai-ris...	AI-011 >> MEAS-3.1	AI-001
12	MAN-3.2	Manage	Change Management	Establish change managem...	Change logs are maintaine...	https://www.nist.gov/ai-ris...	AI-012 >> MAN-3.2	
13	GOV-4.2	Govern	Transparency	Ensure transparency in AI s...	Transparency reports are p...	https://www.nist.gov/ai-ris...	AI-013 >> GOV-4.2	
14	MAP-4.1	Map	Legal Compliance	Identify and comply with a...	Compliance checks are perf...	https://www.nist.gov/ai-ris...	AI-014 >> MAP-4.1	
15	MEAS-4.3	Measure	Monitoring Effectiveness	Monitor the effectiveness o...	Effectiveness reviews are c...	https://www.nist.gov/ai-ris...	AI-015 >> MEAS-4.3	
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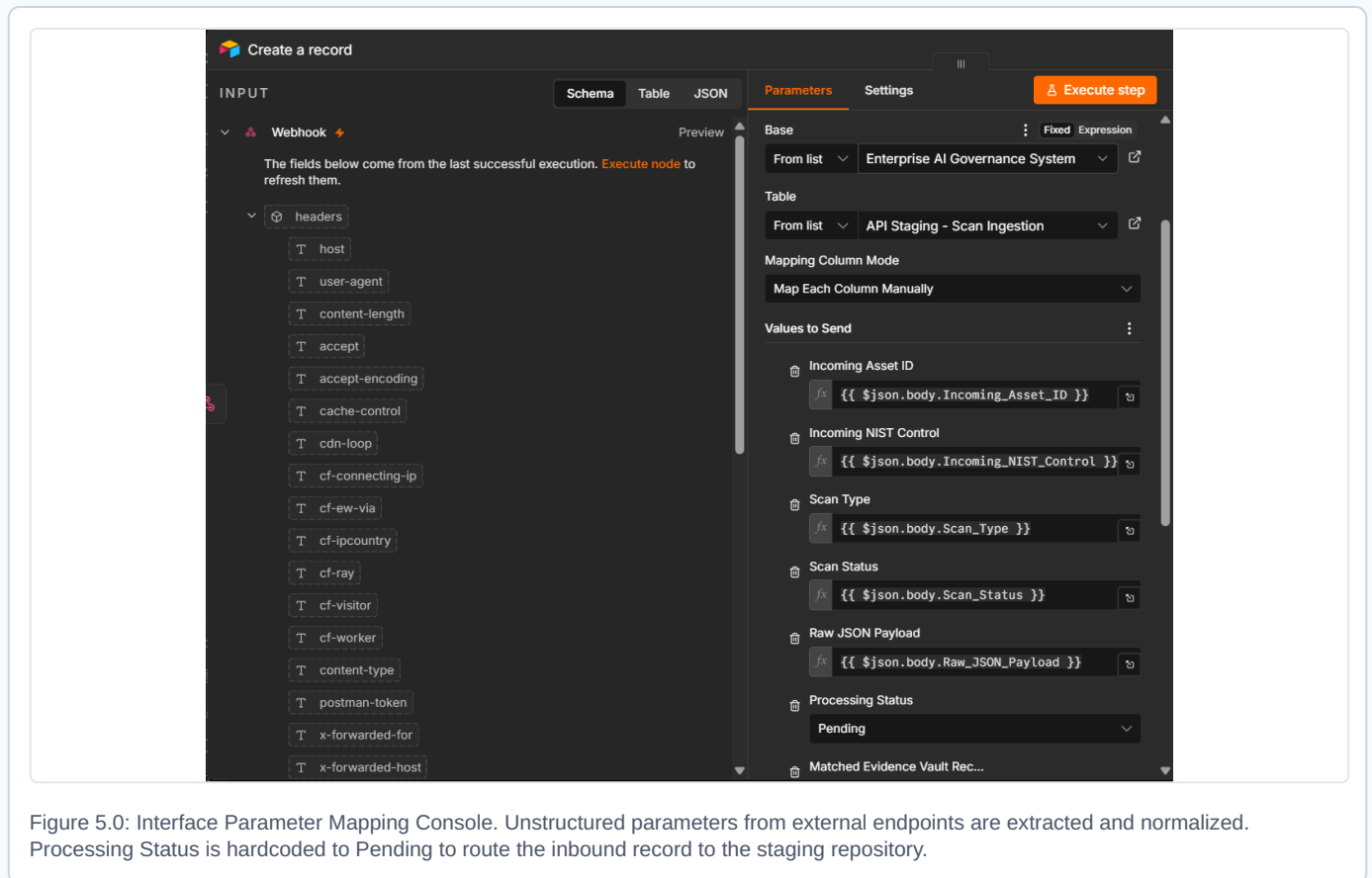
Figure 3.0: NIST AI RMF Core. Regulatory standards and organizational directives are hardcoded into the registry to serve as a standardized benchmark for all subsequent compliance evaluations.

PHASE 2: CONTINUOUS INGESTION AND AI DEFENSE

The Ingestion Gateway



API Staging and Normalization



Autonomous AI Gatekeeper

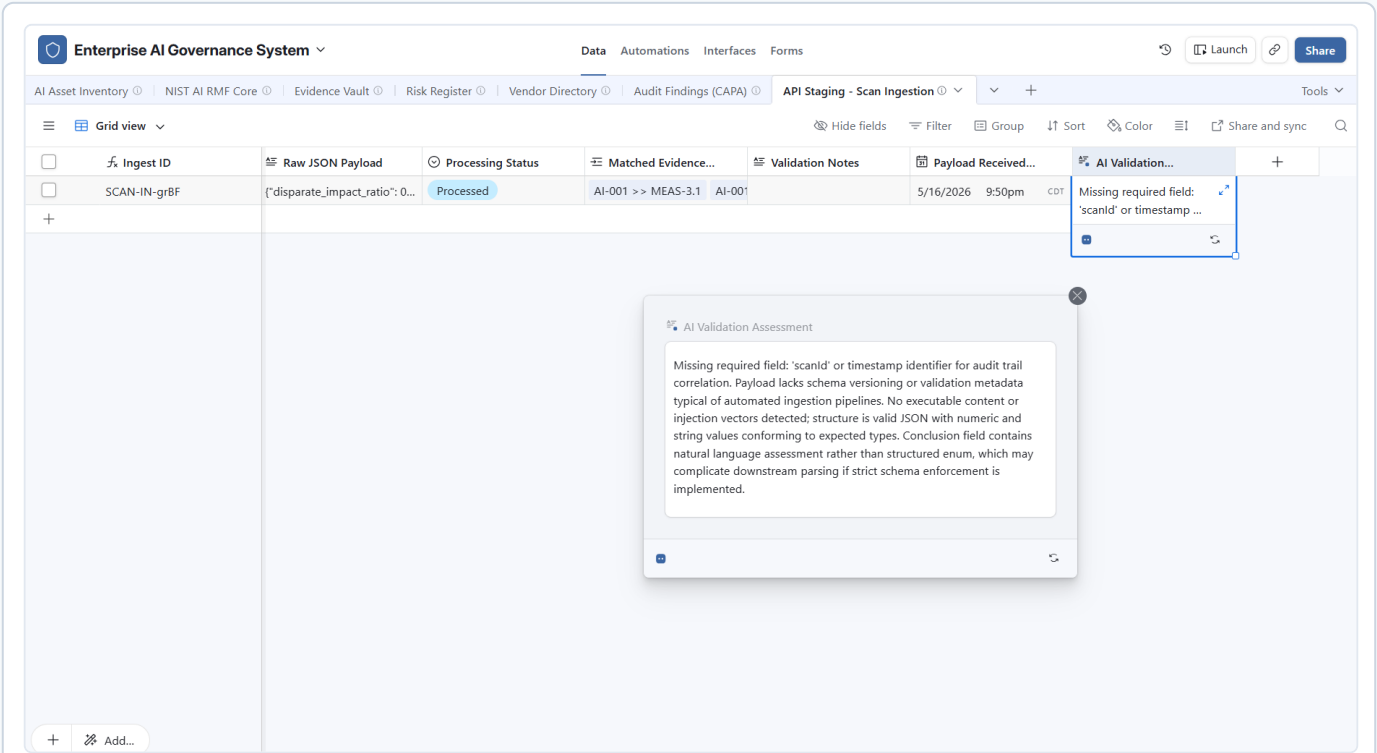


Figure 6.0: Contextual Exception Interception Interface. An automated AI agent checks the payload context. The gatekeeper reads the raw machine payload, checks schema integrity, and writes its assessment notes to prevent non compliant data from passing into production.

PHASE 3: THE RELATIONAL JUNCTION

The Traceability Matrix

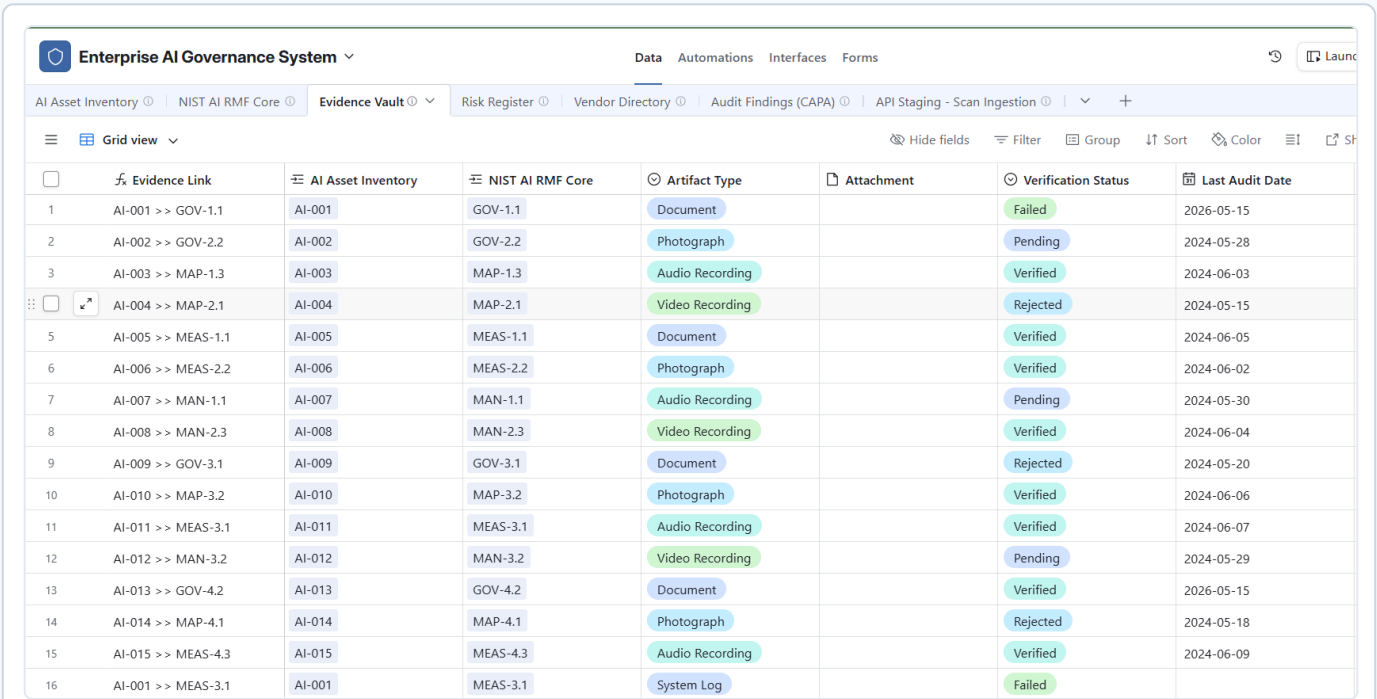
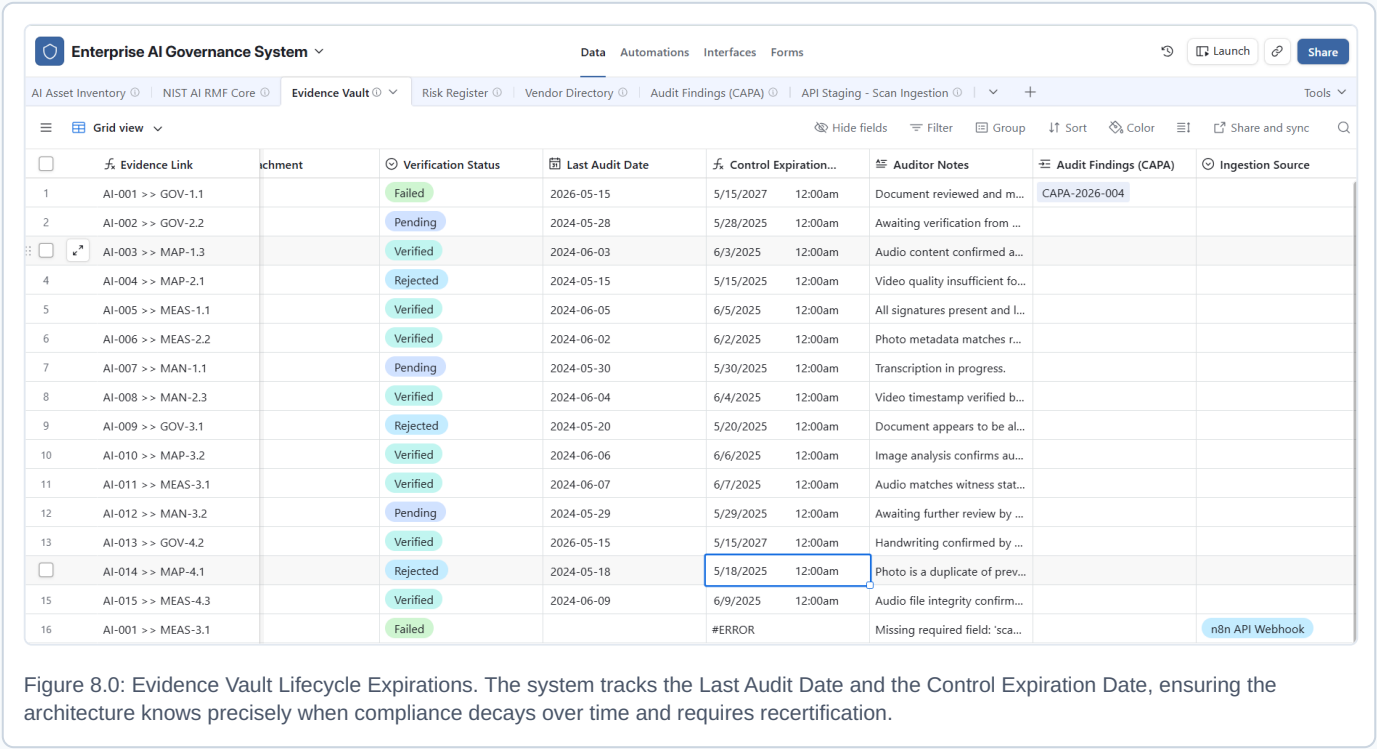


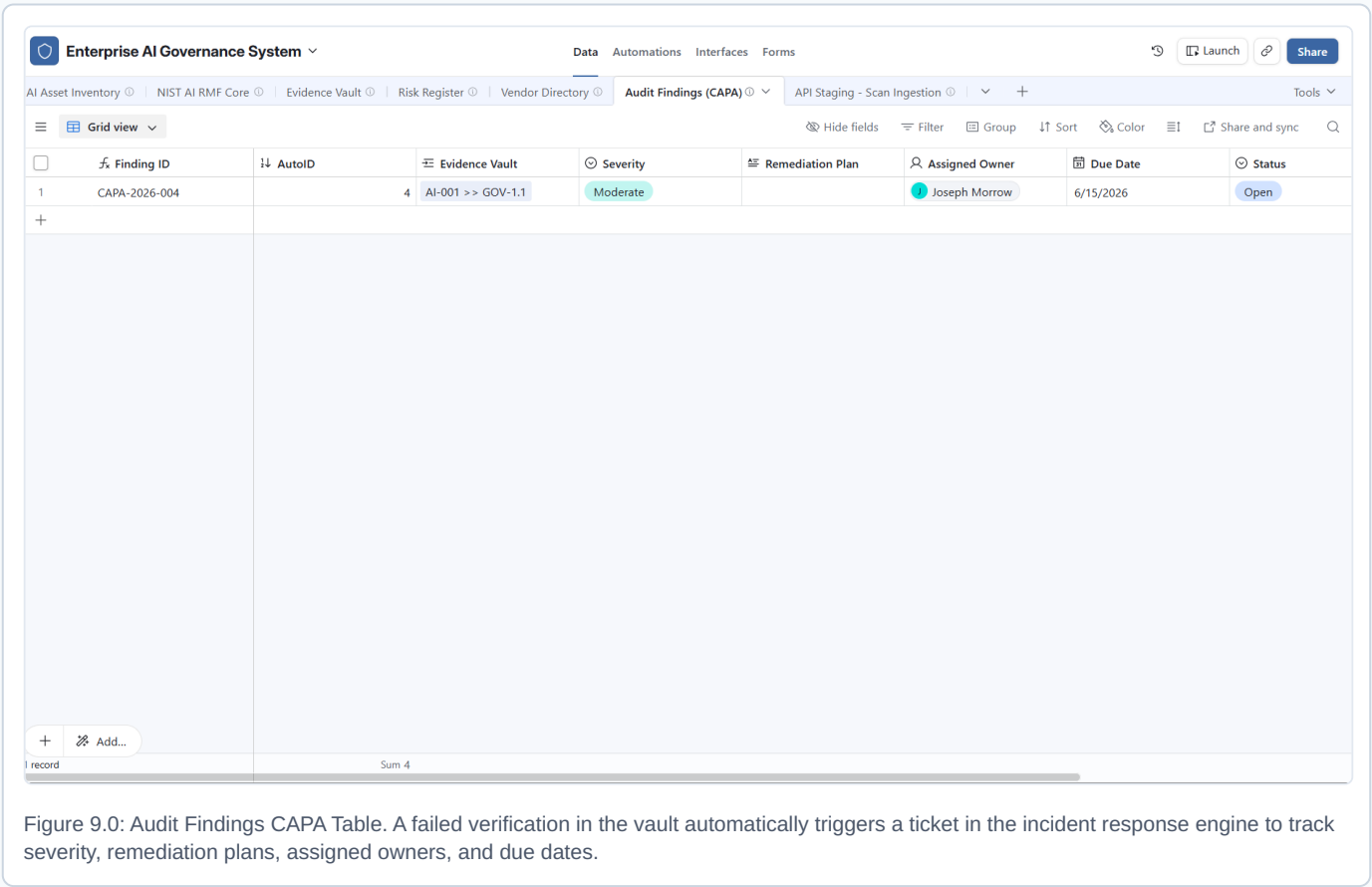
Figure 7.0: Evidence Vault Grid View. The platform utilizes a multi directional link schema operating as a relational junction table. This grid provides an irrefutable mapping that traces an individual asset directly to its associated NIST criteria and verification status.

Control Decay and Expiration



PHASE 4: INCIDENT RESPONSE AND RISK TREATMENT

Automated CAPA Generation



Enterprise Risk Quantification

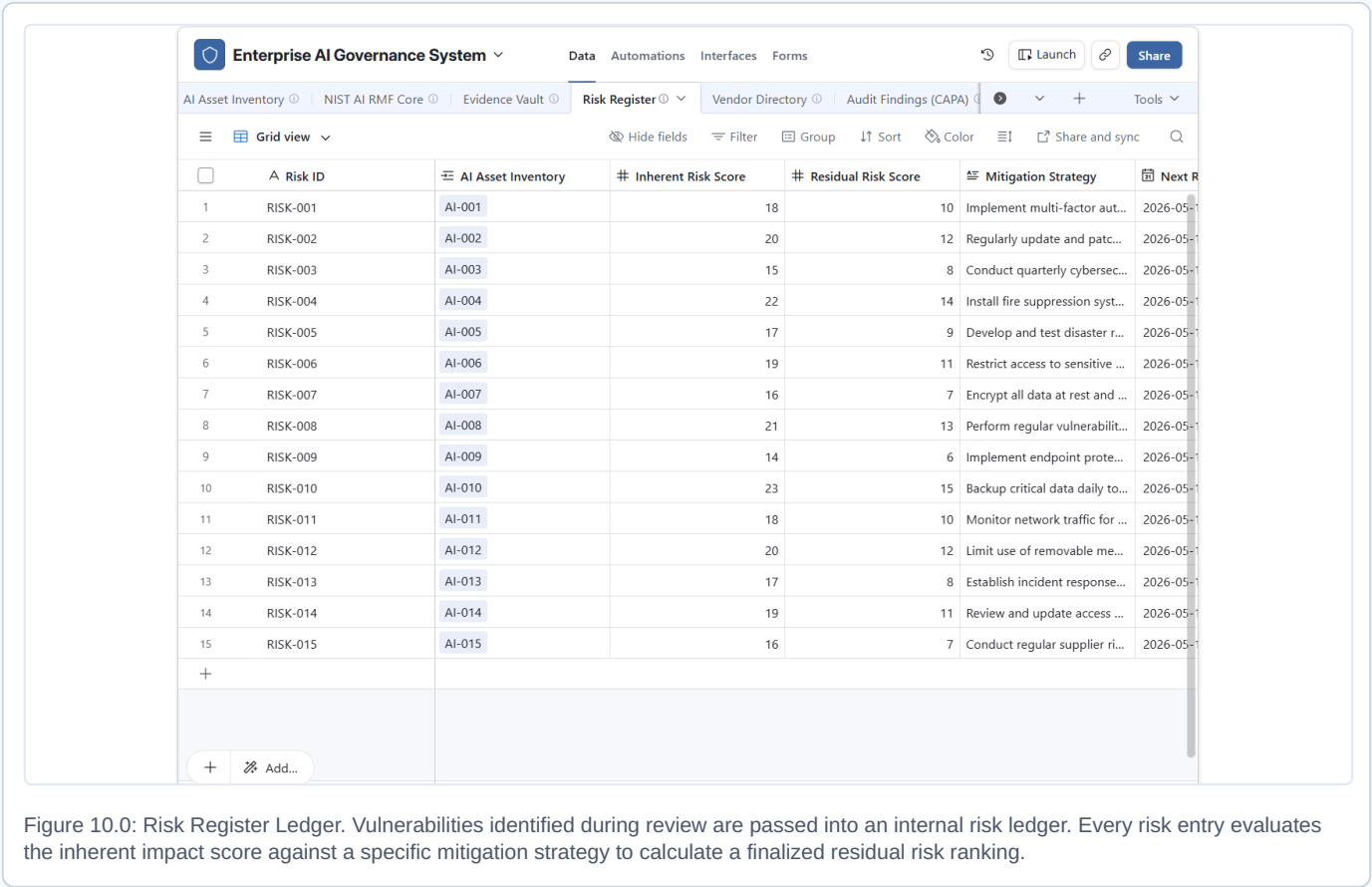


Figure 10.0: Risk Register Ledger. Vulnerabilities identified during review are passed into an internal risk ledger. Every risk entry evaluates the inherent impact score against a specific mitigation strategy to calculate a finalized residual risk ranking.

PHASE 5: EXECUTIVE OVERSIGHT

Portfolio Risk Dashboards

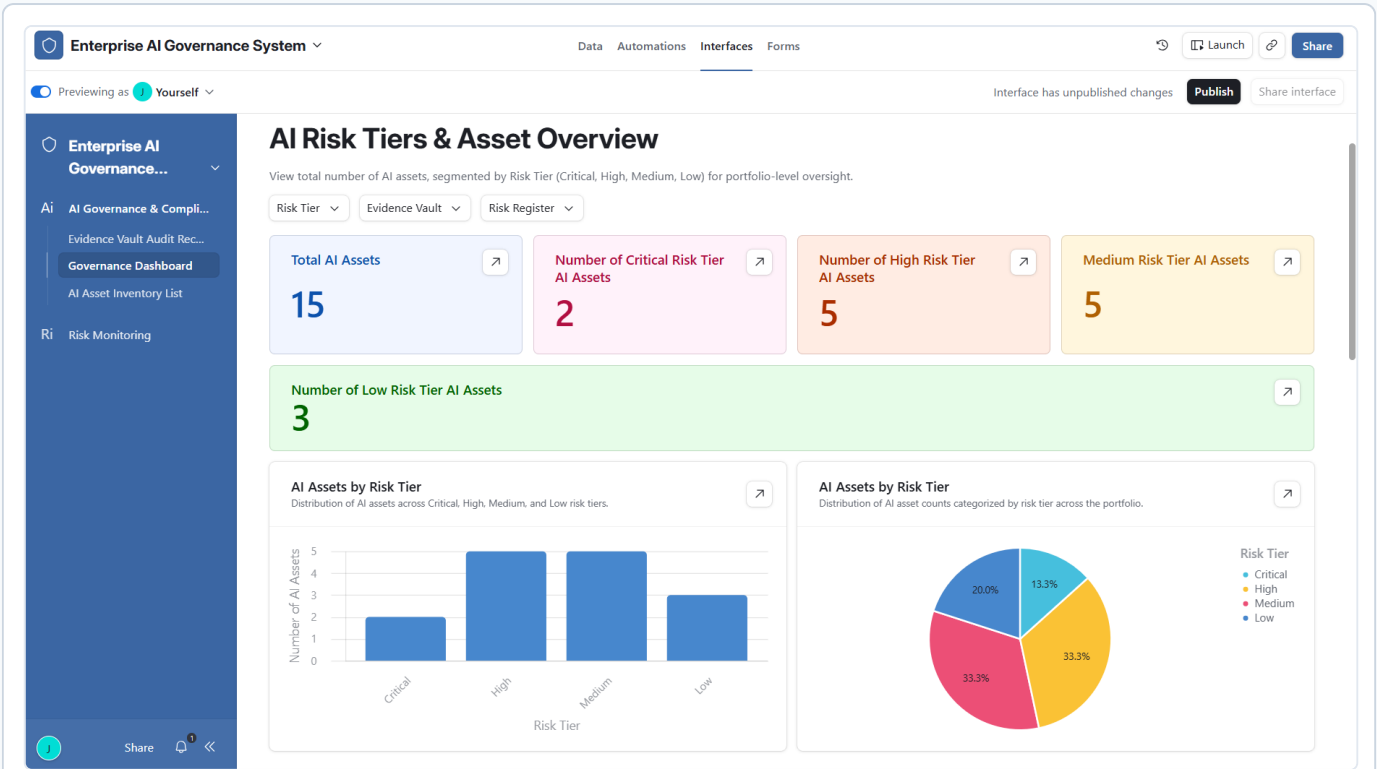


Figure 11.0: AI Risk Tiers and Asset Overview. The platform aggregates granular asset properties to populate executive dashboards, giving leadership immediate visibility into the concentration of high and critical risk models.

Compliance Verification Distribution

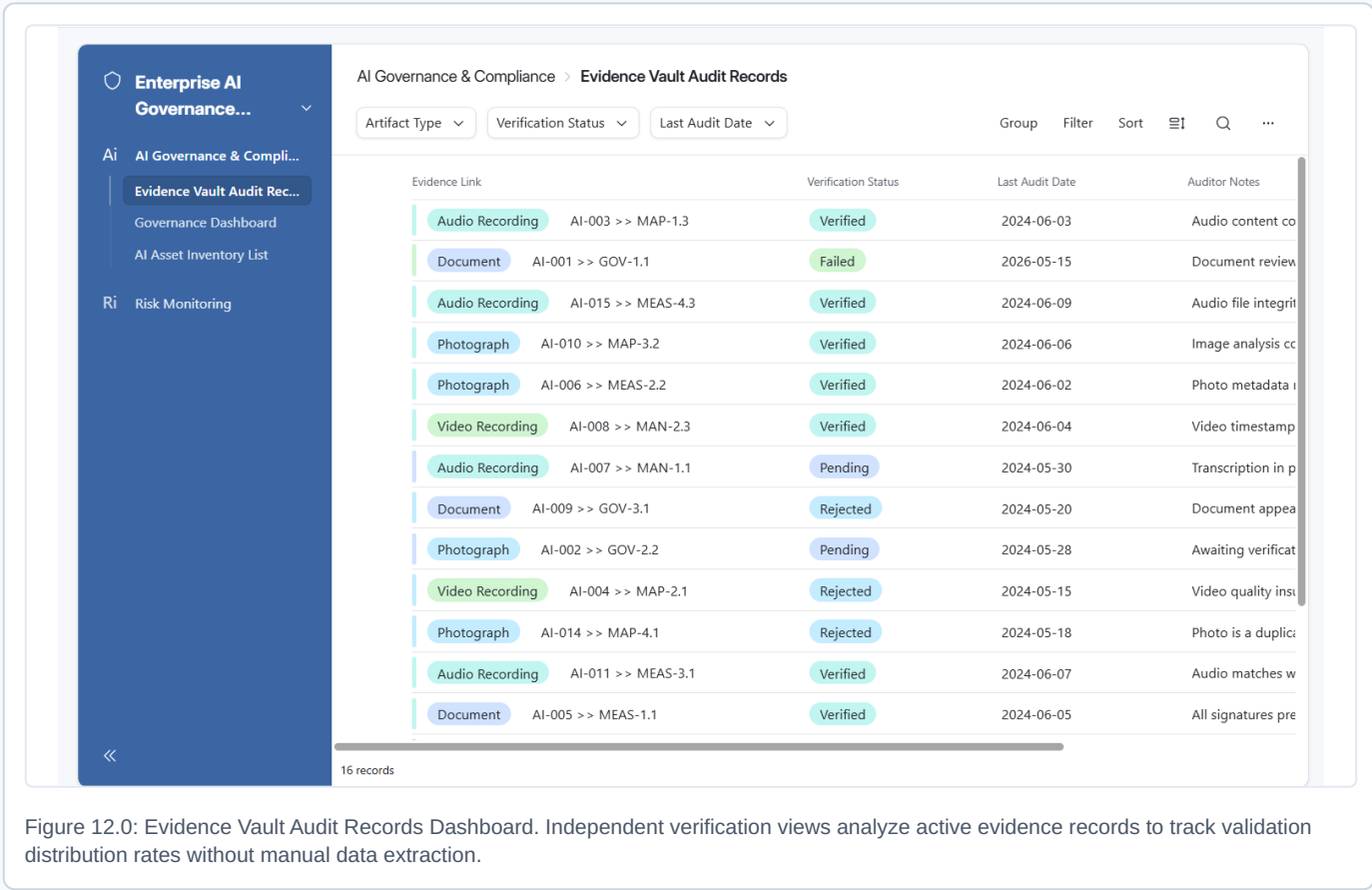


Figure 12.0: Evidence Vault Audit Records Dashboard. Independent verification views analyze active evidence records to track validation distribution rates without manual data extraction.